

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI
DEPARTMENT OF COMPUTER SCIENCE AND INFORMATION SYSTEMS

COMPUTER PROGRAMMING (CSF111)
1ST SEMESTER 2021-22 COMPREHENSIVE EXAMINATION
PART A - CLOSED BOOK

MAX MARKS: 60
TIME: 1 HOUR 50 MINUTES

Instructions:

1. Answer all the parts of a question together in the answer sheet. Answers at separate places will not be accepted.
2. Please do not write any information irrelevant to the question.

1. Is CPU directly connected to the secondary memory? Provide your answer with reference to any one of the two points mentioned below. [4 M]
 - a. If yes, how are the contents of secondary memory accessed?
 - b. If not, provide a justified reason for not having the secondary memory with the CPU.

2. Write a **recursive function** to find an element **key** in a linked list. Assume regular structure definitions of a linked list to contain: (**head**: pointer to the first node, **int size**); and a linked list node to contain: (**next**: pointer to the next node, **int value**). Fill in the code for the find function. This function should print a message "**element found**" if **key** exists in the list; "**element not found**" otherwise. [4M]

```
void findInLL(struct linked_list* list1, int key) {  
    find(list1->head, key); }  
void find(struct node * n1, int key) {  
    // YOUR CODE GOES HERE  
}
```

3. A system stores three floating point numbers in A, B, and C registers. The numbers are stored in IEEE-754 single precision floating point format. The 32-bit content stored in A, B, and C registers are 0xC1400000, 0x42100000, and 0x41400000 respectively. You are required to analyze the pseudocode below and write the correct output. [6 M]

```
if (C = A+B)  
    Output "Condition 1 is true"  
else if (A+C ==0 && B = 3C)  
    Output "Condition 2 is true"  
else  
    Output "None of the conditions is true"
```

4. Consider an accounting system that does all of its computation in 2's complement form. While interacting with the system, let a user give two numbers A and B as 11101101 and 11100110 respectively. The system computes their difference by subtracting B from A and multiplies resulting value with 00001001. Write the final value in 8-bit binary form that should be received by user. [6 M]
5. The below program reads the order of a lower triangle matrix, M first. It then reads the non-zero elements of M row-by-row starting with the first row and stores these elements in one dimensional array "arr" as follows: non-zero element of the first row of M followed by the no-zero elements of the

second row of M and so on. Then, it finds the transpose of M and stores the elements of M^T in "arr1" as follows: non-zero elements of first row of M^T followed by the non-zero elements of the second row of M^T and so on. For example, if matrix A (shown below) is the input matrix, then it is stored in one dimensional array as shown in Figure 1 and the transpose of matrix A should be stored as shown in Figure 2. To achieve the task explained above, write the missing parts/statements of the program without adding any new variables or functions. [10 M]

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 \\ 4 & 5 & 6 & 0 \\ 7 & 8 & 9 & 10 \end{bmatrix}$$

Matrix A

arr:

1	2	3	4	5	6	7	8	9	10
---	---	---	---	---	---	---	---	---	----

Figure 1

arr1:

1	2	4	7	3	5	8	6	9	10
---	---	---	---	---	---	---	---	---	----

Figure 2

```
#include <stdio.h>
#define N 15
int main(){
    int n;
    printf("Enter the order of the matrix"); scanf("%d", &n); int size = 0;
    for(int i=1;i<=n;i++){
        size += i;
        printf("%d\n", size);
        int arr[N] = {0};
        for(int i=0;i<size;i++){
            scanf("%d", &arr[i]);
            int j=0, count = ?, temp = 1, index = 0;
            int arr1[N] = {0};
            while(1){
                int i = j, temp1 = ?, count1 = 0;
                while(1){
                    STATEMENT 1;
                    index++;
                    i = i+temp1;
                    STATEMENT 2;
                    if(temp1 > n)
                        STATEMENT 3;
                }
                STATEMENT 4;
                ++count;
                if(index == size)
                    STATEMENT 5;
            }
            for(int i=0;i<size;i++)
                printf("%d, ", arr1[i]);
            return 0;
        }
    }
```

6. The program given below reads a 2D matrix row-by-row, starting with the first row and finds the largest sum of consecutive elements in that matrix. Assume that such a largest sum could exist either in a row or a column. If a set of consecutive elements in row "i" accumulates to the largest sum, then this program prints "ROWS: " followed by the value of "i", followed by the starting and ending indices of the elements that accumulates to the largest sum in i^{th} row, followed by the largest sum. Similar output is generated if a column has the largest sum by printing "COLUMN: " in place of "ROW: ". For example, consider the

below given 2-dimensional matrices A and B. For matrix A, the program should output "row: 2,0,3,42". For matrix B, the program should output "column: 3,0,3,30". To achieve the task explained above, write the missing statements of the program without adding any new variables or functions. [10 M]

1	2	3	4
5	6	7	8
9	10	11	12
1	2	3	4

Matrix A

1	5	-9	1
2	6	-10	2
3	7	-11	3
4	8	-12	24

Matrix B

```
#define order 4
void main(){
    int arr[order][order];
    for(int i=0;i<order;i++)
        for(int j=0;j<order;j++)
            scanf("%d",&arr[i][j]);
    int is_row = -1, index = -1, start=0, end=0, sum = 0;
    for(int i=0;i<order;i++){
        for(int j=0;j<order;j++){
            STATEMENT 1
            for(int k=j;k<order;k++){
                STATEMENT 2
                if(sum1 > sum){
                    index = i;
                    STATEMENT 3
                    sum = sum1;
                    STATEMENT 4
                    end = k;
                }
            }
        }
    }
    for(int i=0;i<order;i++){
        for(int j=0;j<order;j++){
            STATEMENT 5
            for(int k=j;k<order;k++){
                STATEMENT 6
                if(sum1 > sum){
                    index = i;
                    STATEMENT 7
                    sum = sum1;
                    STATEMENT 8
                    end = k;
                }
            }
        }
    }
    if(is_row)
        STATEMENT 9
    else
        STATEMENT 10
}
```

7. Briefly answer the following:

[4*3 = 12 M]

- Given a machine with the size of unsigned long long as 8 bytes. What should be the range of values for a variable declared with this data type? Also write the format specifier for the same.
- Mention at least 4 techniques to minimize stack usage in a C program.
- What is a memory leak? What are the consequences of a memory leak? Explain with an example.
- Assume that the starting address of an array A[5][5] is 504 and size of each array element is 4 bytes. What is the address of A[4][3], if A is stored in row major form and column major form?

8. Write the output of the following code segments.

[4*2 = 8 M]

a.

```
int a = 0, b = 1;
int c = a || --b;
int d = ++a && b++;
printf("a = %d, b = %d, c = %d,
d = %d", a, b, c, d);
```

b.

```
int x = 2, y = 0;
int z = --x || ++y;
int k = (y++ || ++x) && x--;
printf("x = %d, y = %d, z =
%d, k = %d", x, y, z, k);
```

c.

```
int a = 1;
for(int i = 0; i < 3; i++){
    int a = 0;
    static int b = 0;
    a += 10; b += 5;
    printf("a = %d, b = %d \n", a,
b);
}
```

d.

```
void main(){
    static int a = 10; int b = 5;
    if (b++ && --a){
        printf("%d ", a);
        main();
    }
}
```

[----- END OF PART A -----]

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COMPUTER PROGRAMMING (CSF111)
1ST SEMESTER 2021-22 COMPREHENSIVE EXAMINATION
PART B - OPEN BOOK

MAX MARKS: 40
TIME: 70 MINUTES

Instructions:

1. Answer all the parts of a question together in the answer sheet. Answers at separate places will not be accepted.
2. Please do not write any information irrelevant to the question.

1. Given a book store brand (say "BS") that has outlets in multiple cities of India. We are creating a database (or a catalog) to store details of all BS Stores. Each BS store has the following details that needs to be stored: **storeID**, **city**, **state**, **booksList** (A linked list of books available in that store), **count** (number of books available in that store). **state** can only take the following values **AP**, **UP**, **RJ**, **MH** and **DL**. **city** can be any string. Each book has the following details: **bookID**, **title**, **author**.
 - a. Create Structures for representing a BS store (**struct store**) and a book (**struct book**). Within declaration ensure that the **state** of a store can only take the above-mentioned values. [6M]
 - b. Write a function **struct store * createCatalogOfStores(int n)**, that takes the number of stores **n** as the input parameter and creates an array for storing details of **n** number of stores. The array must reside on the heap. This function should then take **n** sets of inputs of (**city**, **state**) from the user and populate the above created array with this information. The **count** of books for each store can be initialized to zero and the **booksList** for each store can be initialized to **NULL**. You can assign any integer values to **storeID** such that no two stores in the entire catalog have the same value of **storeID**. This function should return the array created. [8M]
 - c. Write a function **struct book * createBook()**, that takes the information of **title** and **author** from the user, creates a new book on the heap and returns it. You can assign any integer values to **bookID** such that no two books in the entire catalog have the same value of **bookID**. [4M]
 - d. Write a function **addBook(struct book * b, int sid)**, that takes a book and a store id, and inserts the book into the store with the given store id. Assume that **booksList** points to the first book in the list (if **count >= 1**). [4M]
 - e. Write a function **findBookInAllStores(int bid, struct store * storeCatalog)** that finds whether a given book with **bookID=bid** is available in any of the stores in **storeCatalog** or not. It prints the message "Book found in the store located in <city>" if the book is found, "Book not found!" otherwise. [6M]
 - f. Write a function **sortBooksOfAllStores(struct store * storesCatalog)** that sorts the **booksList** associated with each store in the **storeCatalog**. Sorting must be performed based on **bookID**. [12M]

[----- END OF PART B -----]